Building a unified science of sensorimotor control with Densely Overlapping Measurement Environments [DOMEs]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. Mission

> Develop and disseminate a new class of scientific instrument called a **Densely Overlapping Measurement Environment (Dome)** that records the complete perception–action loop of a behaving agent as a single calibrated, uncertainty-tagged measurement, and build the autonomous organization dedicated to the measurement itself and form the boundary object that will align perceptual and motor neuroscience, musculoskeletal biomechanics, mobile robotics, and embodied AI into a single convergent science of sensorimotor control.

1.1. A New Observable: The Sensorimotor Loop

Every agent, living or engineered, must solve the same problem - sense a thin slice of available energy in the environment through a limited set of imperfect transducers and on its basis push its body against the world to propel itself towards its goal. Information flows in, forces flow out; the brain exists to yank the bones around.

We can measure individual components of this loop with extraordinary precision progressing at the rate of our technology. Camera and IMU based motion capture can track the kinematics of the body, force plates measure the kinetic forces between the body and world. Outward facing cameras can measure the world, and inward facing cameras can track the eyes. For humans, EEG nets and surface EMG record course records of patterns of neural activity in the cortex and musculoskeletal system. In animal models, dense electrode arrays resolve the individual spikes of neurons and motor units.

And yet, despite the extradradoinary precision and progress along these individual threads of measurement, access to the coherent whole representing the shared observational context that each of these tools illuminates remains ellusive. Occasionally, a research team will reach accross disciplinary divides and stitch several threads together for heroic integrative studies, but the methods rarely escape the Methods sections of a small cluster of publications spanning roughly the timescale of a single PhD.

[TODO - fold next two paragraphs into 'unmet need' and 'vision' sections below] Each of these teams sought out to answer a question at the edge of their discipline and instead found themselves in the business of tool building. The build these tools because they know the answers to the questions that they seek lie in the output of the integrated tool that does not exist. In an institution that rewards the plating of seeds over the tending of gardens, the long and laborious work of maintaining a novel complex instrument quickly overwhelms the capacity of an academic lab. Eventually the nascent tool collapses under the weight of its own complexity, and what was meant to be a bridge between two domains of study instead fades into an evergrowing pile of academic abandonware.

We need a new organization that exists outside of the siloed domains of the Academia that dedicates itself to the measurement, the development and dissertation of world-class, convival tools and novel skillsets explore the landscape the capacity unlocks. That organization is the **FreeMoCap X-Lab** [FMC-X]. We have already changed the landscape human-focused research by bringing high-quality capture to over 15,000 researchers, clinicians, robotists, and animators across 152 countries. With the support of the NSF X-Labs program, we will extend our mission and change entire landscape of neuroscience, biomechanics and robotics.

1.2. A New Instrument: Building the Dome

We define a Dome as a densely instrumented region of real space engineered (at some aspirational extreme) to record every possible measurable channel of the agent–environment interaction at once. It is the opposite of a space telescope, an empty volume of of lenses focused inward at the mysterious place where the body meets the world.

In practice, individual Domes will comprise a partial subset of available instrumentation, selected based on use-case. However, criticall, **All Domes produce the same output** regardless of the component instruments - a metrologically-grounded record of semantically-coherent cannonical models **partially hydrated** by measurements from the available instruments [TODO - Define/invoke Sensor-Grounded Ontology thing here].

A motion capture recording produces the same reconstructed `Head` A mocap `Skull` is assumed to have `Eyes`, reagardless of whether the subject was wearing an eye tracker to hydrate its Degrees of Freedom. Both a ferret and a human have a `Skull` wqhich maybe approxomated as the same `RigidBody` defining segments of the body of a humanoid robot [Collaborator - SH, CH, JH, GN, AS] *[TODO - This robot stuff is redundant to below, I think. need consolldation].* The reconstructed `RetinalInput` derived from reconstructions of the `Body`, `Eye`, and `Environment` of a human participant running across a rocky field [Ref-Collab-MH/KB] are precisely the same data models as those derived from a `Marmoset` leaping from branch to branch with Neuropixel electrodes emdedded in its `SuperiorColliculus` leaping to platform [Ref-Collab-AH/JY], a `Ferret` chasing a fictive prey with a Miniscope viewer mounted over of its `PosteriorParietalCortex` [Ref-Collab-BS], and a `GuineaFowl` running across a pneumatically actuated platform with EMG-electrodes implanted in its `MedialGastrocnemius` [Ref-Collab-MD].

[TODO - reference specific work building DOME-variants in flagship facility and fan-out of built-to-use variants to collaborator network. Mention targetting specific hardware based on practical and research-grounded needs - target Retinal input good enough to target ephys, needs build better eye tracker (dense sensor array for better eye data in general, and adding unmeasurable DoFs like torsion and accomdation for accurate retinal optic flow(CURL). Need world sensor to map environment, built into head-mounted world sensor array of eye tracker, later expanding to attach that scanner to drones. Auto-calibrating camera sensor array - break practicality plateus of large capture volumes, allow for autonomously driven experiments on ideal camera configureations, and ai-driven autonoumous collaborative experiments by intergrating drone-swarms in the scene (i.e. the drone swarms and camera arrays interact to test different capture configurations, training the drone-swarm control policies, all which preps for integrations into Dome-Mobile later to ground inside-out sensors of IMU-suit and head-mounted world scanner))]

1.3. The New Capability

Because every channel is spatially calibrated, temporally synchronized, uncertainty-traced within a unified sensor-grounded ontology, Dome records drop into modern reinforcement-learning and robotics stacks with minimal reshaping — giving embodied AI the real-world, uncertainty-bounded sensorimotor corpora that safe learning requires and that no current dataset provides.

Internally, each sensor stream is modelable agaist every other stream, both within recordings (modelling eye movements against head movent to model VOR) and across recordings (back-fill old datasets with models of Cyclotorsion after we build a new eye tracker that measures torsion). The data also provides internal automatic erorr signals - downstream estimates of body kinematics fused from IMU- and Camera-mocap can back project onto the the original

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cmaera images, defining a reprojection-error signal that can improve image pose-detection algorithms that are currently stuck on an increasingly long-in-the-tooth [TODO - synonym] COCO dataset.

The same **Sensor-Grounded Ontology** that [...allows us to align outputs for heterogoenous Domes...] makes the instrument legible to a human — explicit entities, explicit relationships, no assumed domain knowledge — makes its records natively legible to a machine: typed, grounded data over which a model can build reliable reasoning chains rather than guessing at unlabeled streams. This cleanliness is not a layer added on top; it is the property that lets a Dome route its live streams and logs through a local model to assist novices, experts, and developers alike, and that lets published results tag back into one shared entity–relationship structure connecting them to the wider body of science.

1.4. Unmet Needs

[TODO - this section is redundant and overlapping with other stuff, and also way too verbose. Can be cut down later] An instrument like this is unmet by existing organizational structures and funding mechanisms for reasons that are structural, not incidental. Building a scientific instrument is **a distinct craft, not a downstream application of domain expertise** — mastery of electrophysiology or biomechanics does not transfer to the engineering of a calibrated, maintainable measurement system, and because that difficulty is invisible from outside the craft, integrated measurement is repeatedly scoped as a student side-project, under-resourced, and abandoned when it proves to be a career of engineering rather than a semester of it. The deeper failure is not that these efforts collapse but that **even their successes do not compound**: a research result and a research instrument are different deliverables, and the second — a one-off tuned to a single rig, undocumented and unmaintained — rarely survives the trainee who built it. The field rewards the demonstration, not the durable artifact, and often cannot tell them apart, so proofs of concept are celebrated as finished tools and the unglamorous engineering that would turn one into an instrument others can trust is never done.

Where durable scientific instruments do exist, they were built by escaping this structure rather than succeeding within it — in institutes engineered for the purpose, or by rare leaders who treated maintenance as first-class against their own career incentives. That escape is the necessary condition. By Conway's Law, an enterprise of domain-named departments, journals, and grants can only produce domain-scoped instruments, reviewed by people who score work inside a single domain — never the long metrology that turns a proof of concept into shared infrastructure. A tool defined by a research domain partitions its users into specialists; a tool defined by a measurement unites everyone who needs that measurement and becomes a commons precisely because it imposes none of their goals. The FreeMoCap Foundation already ran this experiment: scoped to the measurement and refusing to claim a domain, it became a commons of over 15,000 users across 152 countries — but a commons cannot form around abandonware, so **the scoping argument and the engineering argument are the same argument**.

The measurement therefore needs its own organization. The PI left a tenure-track position because the institution could not support this work, and founded the FreeMoCap Foundation as a 501(c)(3) whose primary output is open scientific instruments; the X-Lab's operating model follows directly — scope the deliverable to the instrument, hold full-time career engineers for years so the instrument accumulates the mastery that makes it trustworthy, and fan the science out to a standing network of collaborators who use the tools and feed insight back. This is not a critique offered from the sidelines. We know this failure mode because **we built the thing that does not exhibit it**: a working animal-scale instrument and a global user community, engineered to be installed, run, and extended by people who did not build it — exactly the standard the X-Labs model exists to fund, and one a university appointment cannot sustain.

1.5. The Vision

The vision of this proposal represents an grand acceleration of movement the FreeMoCap Project has fostered since 2021, and there is no upper limit to our ambition. The global spread of the FreeMoCap software shows that the world has a hunger for this kind of tool, and we intend to feed it. We will put a Dome in every research lab, every classroom, every Physical Therapy clinic, animation studio, and athletic facility. We will create field-deployable wearble Dome-Mobile systems that provide lab-quality data of humans and animals in the remote wilderness. We will build them into the SCUBA systems of deep sea welders and into the EVA suits and HAB volumes of the ISS. Every country hospital and retirement home will have an AI-enabled, auto-calibrating, self-analyzing Gait and Posture facility that will have plain language conversations with patients about their own kinematic data. We will build Domes into the sides of moutains to measure the aerodyanmics of hunting hawks, and onto the hulls of shipping boats to understand the mechanics of the dolphins that play in their wake. Every Zoo enclosure will become a 24/7 data stream, with animal tracking models jumpstarted through citizen science and then accelerated through auto-training backprojection pipelines. Previously disconnect specialized results on animal models and niche domains will connect through shared links in a Sensor Grounded Ontology, rather than being scattered across publications in specialized journals. The resulting web of scientific beliefs about neuroscience and biomechanics will trace backwards a shared ontology underpinning a unified science of natural behavior fed by every calibrated Dome, be it a $1,000,000 flagship facility or $100 student kit. so that ever reconstructed behavior embodies the same, eternal question - Would you like to know more?

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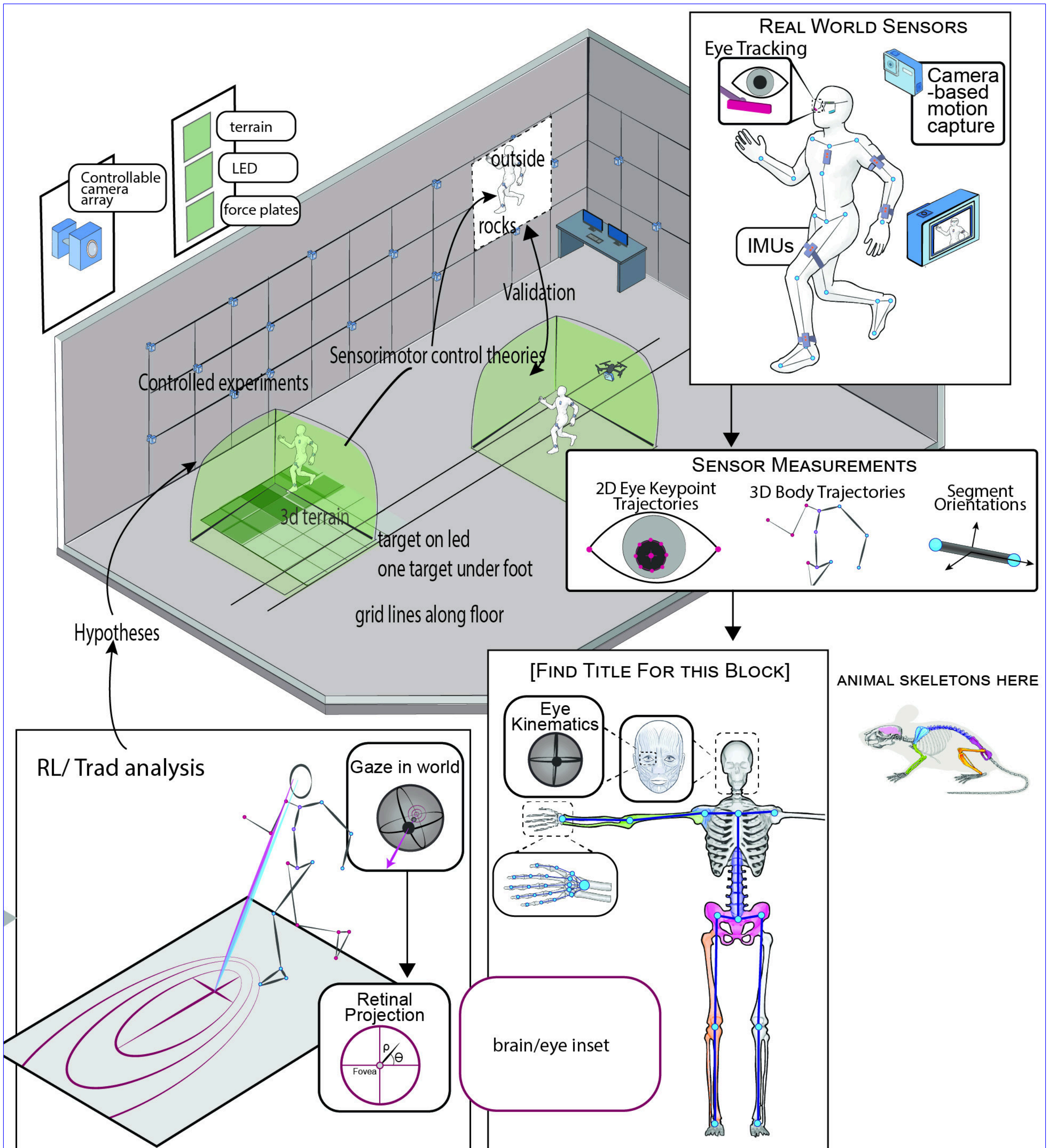

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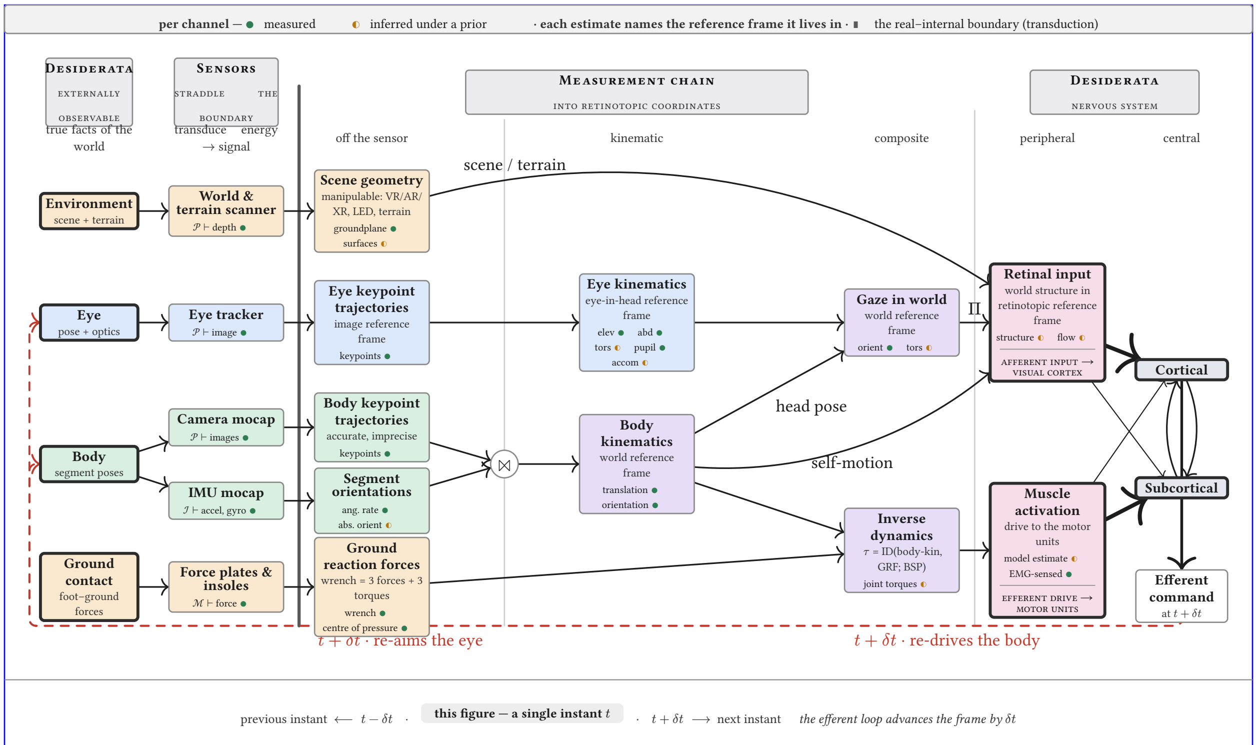

Reading the chart both ends are **desiderata** — the True Facts we want. The chain between them transports each estimate out of its sensor's reference frame and into the reference frame the nervous system uses.

Notation

$\mathcal{E}$	an **energy** a sensor is sensitive to: $\mathcal{P}$ photons · $\mathcal{I}$ inertial · $\mathcal{M}$ contact
$\mathcal{E} \vdash s$	**transduce** — a sensor **yields** ($\vdash$) a raw **signal** s; the one physical step, at the real–internal boundary
$\to$	**derive** — a signal is processed into a measurement, then an estimate
$\oplus$	**compose** — chain estimates across reference frames
$\bowtie$	**fuse** — combine two estimates of one quantity
Π	**project** — map a state onto an observable

The reference-frame journey

Each estimate lives in one reference frame. The vision path is a journey from sensor reference frames into the retinotopic reference frame — the coordinates the visual cortex uses.

Vision	image → eye-in-head → world → retinotopic
Body	camera / IMU sensor → world (inertial)
Gaze	eye-in-head $\oplus$ head-in-world → world

The chains — energy $\vdash$ signal → estimate

Eye	$\mathcal{P} \vdash$ image → keypoints → eye-kin (elev, abd, tors, pupil, accom)
Body	$\mathcal{P} \vdash$ images → keypoints; $\mathcal{I} \vdash$ accel+gyro → segment orient; body-kin = keypoints $\bowtie$ orient
Retinal input	Π(gaze, scene, self-motion) — afferent
Motor	τ = ID(body-kin, GRF); muscle = M(τ) $\bowtie$ EMG — efferent

Figure 2: **The measurement chain of a Dome.** The two ends are **desiderata** — the True Facts we want to know. On the left, the **externally observable** facts (the eye, the body, the environment); on the right, the facts that are **not** externally observable (the afferent retinal input, the efferent muscle activation, and the central nervous system). Between them sit the **sensors**, which straddle the real–internal boundary by transducing physical energy into signal (the heavy vertical rule), and the **measurement chain** that derives estimates from those signals. Its single job is to carry each estimate out of its sensor's reference frame and into the reference frame the nervous system uses: the vision path runs image ⟶ eye-in-head ⟶ world ⟶ **retinotopic**, the coordinates of the visual cortex. Every box is a bundle of **channels**, each marked measured (●) or inferred under a prior (◐), and every estimate names the reference frame it lives in. Fusion ($\bowtie$) and composition ($\oplus$) build the kinematic and composite estimates; projection (Π) maps them onto the retinal input and the muscle activation. Those two peripheral signals drive the **central nervous system** (cortical and subcortical, reciprocally coupled), whose efferent command re-aims the eye and re-drives the body at the next instant — closing the loop. The whole figure is a **single instant t**; the dashed red loop advances the frame by δt. The strip beneath the diagram gives the full reading rule, every symbol defined.

2. Technology Landscape

2.1. The measurement threads and their instruments

- The DOME unifies six measurement threads, each served by a mature but isolated instrument ecosystem.
  - Kinematics: markerless pose estimation (DeepLabCut, SLEAP, VitPose, FreeMoCap), marker-based (Vicon, Qualisys, OptiTrack), and inertial (Xsens, Notch), with a gap between clinically-valid kinematics and inverse-dynamics-grade accuracy.
  - Kinetics: force plates (AMTI, Bertec), pressure-sensing insoles, and instrumented treadmills, with ground reaction forces still hard to measure outside the lab.
  - Eye tracking: Pupil Labs (Core, Neon, Invisible), Tobii, and SMI, none of which capture torsion or accommodation.
  - Neural recording: Neuropixels, miniscopes (1-photon calcium imaging), and mobile EEG, with neural activity still hard to record during unconstrained movement.
  - Muscle activity: surface and intramuscular EMG at motor-unit resolution, with a gap between what OpenSim models need and what current sensors deliver.
  - Environment: photogrammetry, Gaussian splatting, neural radiance fields, RGB-D (stereo, LiDAR, structured IR), and event cameras.

2.2. The composition problem

- These instruments do not compose, for four concrete reasons.
  - Coordinate frames are incompatible, because each sensor defines its own.
  - Clocks are unsynchronized, because there is no common timebase.
  - Semantic schemes are non-interoperable: a “gaze point” in one system is not the same quantity as in another.
  - Uncertainty does not propagate across modality boundaries.
- Mobile Brain/Body Imaging (Makeig et al. 2009) is the fifteen-year case study: a methodology, not a platform, because the integration burden falls on each lab.

2.3. The metrology gap

- The field co-records without co-measuring.
  - Calibration quality $\kappa = (\kappa_s, \kappa_t)$ is the fundamental bottleneck. [Reference observability calculus.]
  - The GUM/VIM framework applied to multimodal capture shows why an unbroken chain of traceable calibrations matters — and why no existing system provides one.

2.4. The ontology gap

- Existing tools are sensor-specific rather than measurement-specific.
  - They are built around a particular camera model or force-plate vendor, not around the measurement itself; kinematics is kinematics whether it came from cameras or IMUs.
  - The consequence: data from different labs, species, and equipment generations cannot be compared, pooled, or cumulatively improved.
  - The DOME’s sensor-grounded ontology is the solution: define the measurement, not the sensor.

2.5. The CV field’s plateau

- Computer-vision pose estimation has drifted away from the pixels toward ungroundable targets.
  - Targets such as 3D-from-2D, SMPL mesh fitting, and semantic segmentation lack physical ground truth.

  - Trackers still benchmark against aging COCO data scraped from early-2000s Flickr, and fail on out-of-distribution movement — clinical populations, extreme athletics, animal locomotion.
  - The field needs physical ground truth at scale, which a calibrated multi-camera array provides, along with corpora of the movements it currently cannot track.

2.6. The robot training bottleneck

- Reinforcement learning for legged robots is limited by the sim-to-real gap, not by simulation.
  - Simulation has advanced rapidly (MuJoCo, Isaac Lab, Genesis), but sim-to-real transfer remains the critical barrier.
  - The missing resource is ground-truth, uncertainty-tagged, multimodal recordings of real agents in real environments.
  - Existing datasets (Human3.6M, AMASS, GRAB) are single-modality, lab-constrained, or lack uncertainty quantification.

2.7. Quantitative landscape figure

- A single figure will locate current capability against the DOME's target.
  - For each modality, plot current commercial and academic capability against the DOME's target, with the "uninstrumented region" highlighted.
  - Axes: spatial precision, temporal synchronization quality, degrees of freedom captured, and usability/deployability.

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3. Outcomes

3.1. DOME-L flagship facility

- DOME-L is a warehouse-scale instrumented volume in the greater Boston area.
  - Specs: configurable capture volume up to [TBD] m³, [N] actuated cameras, a modular force-plate floor array (each 0.5 m² panel a force plate, terrain panel, LED panel, or stacked combination), integrated ARGPv3 (LED floor + wall panels, projection, VR), and adjacent animal and fabrication facilities.
  - Phase 0 deliverable: site selection, facility design, and initial build-out.
  - Phase 1 deliverable: an operational DOME-L able to cross-validate DOME-S and DOME-Mobile.
  - KPI: inter-sensor synchronization jitter, end-to-end spatial uncertainty on a reference object, and reprojection error on a standardized movement corpus.

3.2. DOME-S dissemination

- DOME-S extends FreeMoCap's existing webcam-based capture volumes into the disseminated, lab- and classroom-scale instrument.
  - Phase 0: validate DOME-S against DOME-L on a standardized movement battery, establish a calibration protocol any lab can follow, and publish a build guide and parts list.
  - Phase 1: deploy DOME-S at [N] collaborator sites spanning human biomechanics, robotics, visual neuroscience, and clinical applications.
  - KPI: number of deployed and validated DOME-S instances, kinematic accuracy against the DOME-L reference, and inter-site measurement comparability.
    - SPECIFIC TARGET - 510(k) certification of FreeMoCap and DOME instrument as FDA certified for clinical rehab and assessment

3.3. DOME-Mobile wearable

- DOME-Mobile is the wearable form: an IMU suit, binocular eye tracker, and head-mounted world-camera array (stereo RGB, structured IR, LiDAR).
  - Phase 0: sensor selection, initial integration, and benchtop validation.
  - Phase 1: validate DOME-Mobile against DOME-L on locomotion tasks (walking, running, obstacle navigation, stair climbing), indoors and outdoors.
    - Build Drone Swarm Mocap - Drones mounted with simnilar world-scanner on the eye tracker, used to ground the IMU- mocap and flesh out world-scan from head-mounted scanner.
  - KPI: kinematic accuracy against the DOME-L reference during co-recorded trials, gaze-in-world angular uncertainty, and drift over [TBD]-minute outdoor walks.

3.4. Eye tracker development

- The binocular eye tracker measures torsion and accommodation at 200+ Hz.
  - Phase 0: bench prototype measuring torsion (via iris texture) and accommodation (via higher-order Purkinje reflections) binocularly, validated against a reference eye tracker and an artificial eye.
  - Phase 1: mobile form factor, integrated into DOME-Mobile and DOME-S, validated on human participants during natural locomotion.
  - KPI: torsion accuracy [TBD arcmin], accommodation accuracy [TBD diopters], gaze-in-world uncertainty during walking [TBD degrees], and per-unit cost target [TBD].

3.5. Actuated camera array

- The actuated camera array places cameras on controllable mounts that recalibrate under dynamic reconfiguration.

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  - Phase 0: prototype array of [N] cameras on controllable mounts, a calibration method for dynamic reconfiguration, and validation on static and moving reference objects.
  - Phase 1: production array in DOME-L, a remote configuration API, and auto-calibration on sub-volume selection.
  - KPI: post-reconfiguration calibration quality (κ_s, κ_t), time from sub-volume selection to calibrated capture, and reprojection error against a fixed-calibration baseline.

3.6. Camera↔IMU sensor fusion

- Camera↔IMU fusion combines outside-in and inside-out estimates with explicit per-joint uncertainty.
  - Phase 0: develop the fusion algorithm on existing FreeMoCap + IMU-suit data, validate against the DOME-L reference on standardized movements, and establish uncertainty budgets per joint per movement type.
  - Phase 1: real-time fusion in DOME-L and DOME-Mobile, validated for inverse dynamics (joint torques, muscle forces via OpenSim).
  - KPI: joint-center uncertainty at k=2, joint-torque uncertainty against force-plate ground truth, and muscle-force uncertainty against EMG-informed estimates.

3.7. Reprojection-error training pipeline

- The reprojection-error pipeline turns calibrated 3D reconstructions into a training signal for pose estimators.
  - Phase 0: build a pipeline that back-projects calibrated 3D reconstructions onto each camera's 2D view, computes per-joint reprojection error, and uses that error to fine-tune pose estimators.
  - Phase 1: demonstrate measurable improvement on out-of-distribution movements (clinical gait, acrobatics, animal locomotion).
  - KPI: percentage reduction in pose-estimation error on held-out movements, and improvement on standard benchmarks.

3.8. Robot RL integration

- DOME data feeds robot reinforcement learning and inverse reinforcement learning.
  - Phase 0: define a DOME export format mapping directly to MuJoCo/Isaac Lab rigid-body models, record an initial human-locomotion corpus, and demonstrate policy learning from DOME data in simulation.
  - Phase 1: sim-to-real transfer of DOME-trained policies onto physical robots, inverse RL to extract apparent human control policies, and tests of IRL-derived hypotheses in DOME-L via ARGP perturbation.
  - KPI: sim-to-real transfer success rate, policy performance against a simulation-trained baseline, and IRL reward-function predictive accuracy on held-out behavior.

3.9. ARGPv3

- ARGPv3 is a modular augmented-reality ground plane extending the PI's published ARGPv1 apparatus (Matthis 2013–2017).
  - Phase 0: design a modular LED floor-panel system (0.5 m² tiles), prototype with projection-based visual stimuli, and validate that visual perturbations produce measurable locomotor adjustments in DOME-L.
  - Phase 1: full DOME-L floor integration, VR-headset integration for immersive manipulation, and coupled LED floor + wall panels for full visual-field control.
  - KPI: latency from perturbation command to visual update, spatial calibration between LED panels and the DOME coordinate frame, and magnitude of locomotor adjustment per unit visual perturbation.

3.10. Cross-species validation network

- The cross-species network deploys functionally equivalent instruments at animal-collaborator labs.
  - Phase 0: establish calibration and data-exchange protocols with ferret (visual neuroscience + ephys), mouse (systems neuroscience + miniscope), guinea fowl (musculoskeletal biomechanics + EMG), and marmoset (primate electrophysiology) labs.
  - Phase 1: deploy functionally equivalent DOME-S instances at each site and validate cross-species measurement comparability.
  - KPI: number of validated cross-species measurement channels, and inter-species kinematic-model alignment quality.

3.11. Milestone timeline

- The milestones compound across phases, from components to an operating network.
  - Phase 0 (9 months): DOME-L site selection and design, eye-tracker bench prototype, actuated-array prototype, camera↔IMU fusion algorithm, reprojection pipeline v1, DOME-S validation protocol, and collaborator MOUs.
  - Phase 1 Go/No-Go (month 7–9): demonstrated components, a validated calibration chain, and signed collaborator agreements.
  - Phase 1 (24–36 months): operational DOME-L, validated DOME-S at collaborator sites, validated DOME-Mobile, mobile eye tracker v1, actuated array in DOME-L, real-time fusion, robot RL corpus v1, ARGPv3 operational, and cross-species data exchange operational.

NOTE - Names abbreviated in public draft. Check initials to real names in local .env (or request it, if you need it)

3.12. Jonathan Matthis, PhD — President / PI

- Jonathan Matthis (President/PI) founded and maintains FreeMoCap and has spent two decades measuring the full perception–action loop.
  - Left a tenure-track position (Assistant Professor, Human Movement Neuroscience, Northeastern University) specifically because the institution could not support the integrated tool-building this work requires.
  - NEI NIH K99/R00 recipient; 1,400 citations.
  - Published across the loop: gaze and gait during outdoor locomotion (Matthis et al. 2018, Current Biology); reconstruction of retinal input during real-world behavior (Matthis et al. 2022, Current Biology; Muller et al. 2023, Scientific Reports); and the augmented-reality ground-plane paradigm (Matthis et al. 2013, 2014, 2015, 2017).
  - Built and maintains the open-source tooling and the global community the DOME extends.

3.13. NR — CEO

- NR (CEO) brings technology-sector chief-executive experience, freeing the PI to focus on the technical Mission. [Details to fill from .env]
  - Previously founded and ran their own business.
  - Hired as CEO so the PI can concentrate domain expertise on the core technical work rather than organizational operations.

3.14. EI — CTO

- EI (CTO) brings decades of enterprise software architecture — expertise largely absent from academic contexts. [Details to fill from .env]
  - Chief architect and CTO for established software companies.
  - Will ensure world-class enterprise-grade architecture in an open-source project, and build training and workshop infrastructure to inject professional software-development practice into the scientific community.

3.15. AC, PhD — Chief Scientific Officer

- AC (CSO) validated FreeMoCap against research-grade optical motion capture and leads clinical dissemination. [Details to fill from .env]
  - Dissertation validated FreeMoCap against Qualisys, demonstrating clinically valid kinematics from commodity hardware.
  - Expertise in clinical research-tool validation, bench-to-bedside development, neuroprosthetics design, and brain imaging.
  - Core FreeMoCap developer and validation lead; holds the skillset for DOME-S development and clinical dissemination.

3.16. JKL, PhD — Chief AI Officer

- JKL (CAIO) deploys AI systems across classroom, laboratory, and enterprise environments. [Details to fill from .env]
  - PhD in Cognitive Science; Director of the Cognitive Science Program; currently Senior AI Applications Engineer at [SOLID].
  - Will build internal AI systems for development assistance, user support, and educational deployment.

3.17. RR — CFO

- RR (CFO) runs financial operations and NSF grant management. [Details to fill from .env]
  - Bookkeeping, NSF grant management, and financial operations; founded their own bookkeeping cooperative.

  - Taught finance theory at the university level; ensures organizational health and entrepreneurial development.

3.18. KM — Project Manager / Mobile DOME Lead

- KM leads DOME-Mobile and drove the PI's outdoor retinal-optic-flow work. [Details to fill from .env]
  - Key driver of Muller et al. 2023 (retinal optic flow during outdoor locomotion); worked with the PI on core technology for Matthis et al. 2022.
  - Holds the sub-skills to build DOME-Mobile's core instrumentation.

3.19. MN — Project Manager / DOME-L Lead

- MN leads DOME-L and brings clinical-biomechanics and large-lab systems expertise. [Details to fill from .env]
  - Expertise in clinical biomechanics, medical-systems design, mechatronics for clinical tools, and multi-modal motion-capture lab management.
  - Holds the skillset for large-scale DOME-L facility design, construction, and operation.

4. Team Capabilities Statement

4.1. Complementary expertise

- The leadership team spans the full stack the DOME requires, a combination no single institution's faculty could assemble.
  - Scientific domain expertise: JSM (perception/action neuroscience), AC (clinical biomechanics).
  - Enterprise software architecture: EI.
  - AI systems: JKL.
  - Hardware and mechatronics: KM (mobile/wearable), MN (large-scale facilities).
  - Organizational operations: NR (CEO), RR (CFO/finance).
  - The combination exists because the FreeMoCap Foundation was built from the ground up to produce integrated scientific instruments.

4.2. Governance during Phase 0 and Phase 1

- The FreeMoCap X-Lab (FMC-X) will operate as an independent project within the FreeMoCap Foundation, a 501(c)(3) nonprofit.
  - The Foundation runs other work (e.g., the ongoing FreeMoCap Project), but the missions almost fully overlap.
  - Leadership is nearly identical between FMC-F and FMC-X, with final decision authority resting with the President/PI at the top of the org chart.
  - Research, partnership, and organizational decisions are made in days, not weeks, with no higher management to consult.

4.3. Autonomy

- The FreeMoCap Foundation is already autonomous as of this submission.
  - Full internal control of funding allocation, research direction, partnership agreements, intellectual property, and hiring, with no parent-institution approval required for any operational decision.
  - Satisfies all conditions of the NSF X-Labs Autonomy Factor Test (§6.1.1) at the time of submission.

4.4. Collaborator network

- A standing, active network of research groups already spans every domain the DOME serves.
  - Members include human perception-and-action researchers, robotics and prosthetics groups, visual neuroscientists studying ferret and mouse, musculoskeletal biomechanists studying guinea fowl, and primate electrophysiologists.
  - The network is real and active, not aspirational, with existing collaborations and shared tooling across multiple sites.
  - The PI architects and holds the instrument-plus-network together; the domain arms are executed by named collaborators.

4.5. Community governance and engagement

- The community sets design and roadmap priorities through structured processes that gather input without stalling on consensus.
  - A Request-for-Comments process modeled on Python's PEP system collects broad input on design decisions, work targets, and roadmap priorities.
  - Semi-annual in-person convening: Workshops (train students and out-of-domain experts), Hackathons (onboard developers and connect normally remote workers), Conferences (share DOME-based research), and a Congress (organizational status, plans, and direction).

    - A Community Grants Program and Developer Fund support the open-source network and serve as a testbed for engagement mechanisms (Eurovision-style voting, DAO mechanisms, and others).

4.6. FreeMoCap as completed pilot

- FreeMoCap has already proven this model at smaller scale.
    - It is an open-source markerless motion-capture instrument, built and maintained outside academia, scoped to the measurement — turning cameras and light into calibrated, usable skeleton estimates with an obsessive focus on usability — and refusing to claim any research domain.
    - That refusal is what made it a commons.
    - Receipts: over 15,000 users from 152 countries, over 10,000 GitHub stars, and over 3,500 Discord members.
    - The DOME applies this proven model — measurement-scoped, master-built, open, full-time-maintained — to the far larger target of the complete interaction loop, at a scale only dedicated, autonomous, milestone-based X-Labs funding can reach.

4.7. IP management and dissemination strategy

- The IP strategy keeps the core instrumentation open while actively pursuing adoption and commercialization. [To develop — OT contract requires an IP Management Plan by end of Phase 0.]
    - Open-source core instrumentation with permissive licensing for downstream use.
    - IP structured so platform technologies stay widely accessible while the team pursues adoption, commercialization, and ecosystem growth.

516 Bibliography

DRAFT - Work in Progress